

MODELING ROBOTS

Fundamentals of Robotics
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Why modelling

- Our purpose in robotics is how to endow a mechanical machine with some degree of intelligence
- A robot's intelligence is expressed through the physical interaction with the surrounding world
- Thereby it is crucial to understand what a robot can do with its own body
- As humans, we have the same problem.
- We learn our body and how to use it *via* our experiences since the first days of our lives

Why modelling

- Can a robot do the same?
- Well, in some cases, using experiences and learning to figure out how to transfer intent into action is possible and useful.....
- However, most of the times we know quite a bit of the physical structure of a robot
- This knowledge comes from physics and is expressed by a mathematical model
- Mathematical models can be manipulated by mathematical means with *guaranteed* results for control and motion planning.

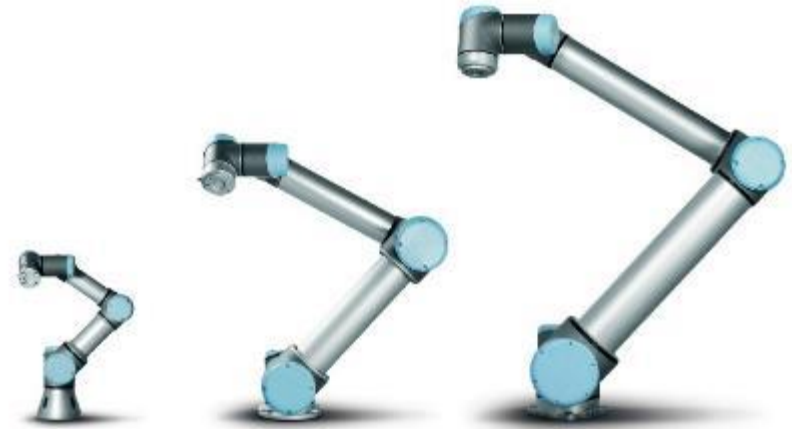
Modelling what?

- We will now spend a few hours to understand the math of robotic modelling
- We will try to see how to capture geometry, kinematics and dynamics using a quite general mathematical formalism
- Do not be scared!!!!
- We will not dig too deep
- All we need to understand is how to create and use model in simulation and design

Some taxonomy and terminology

- A first important distinction is between:

MANIPULATORS: robots with a fixed base



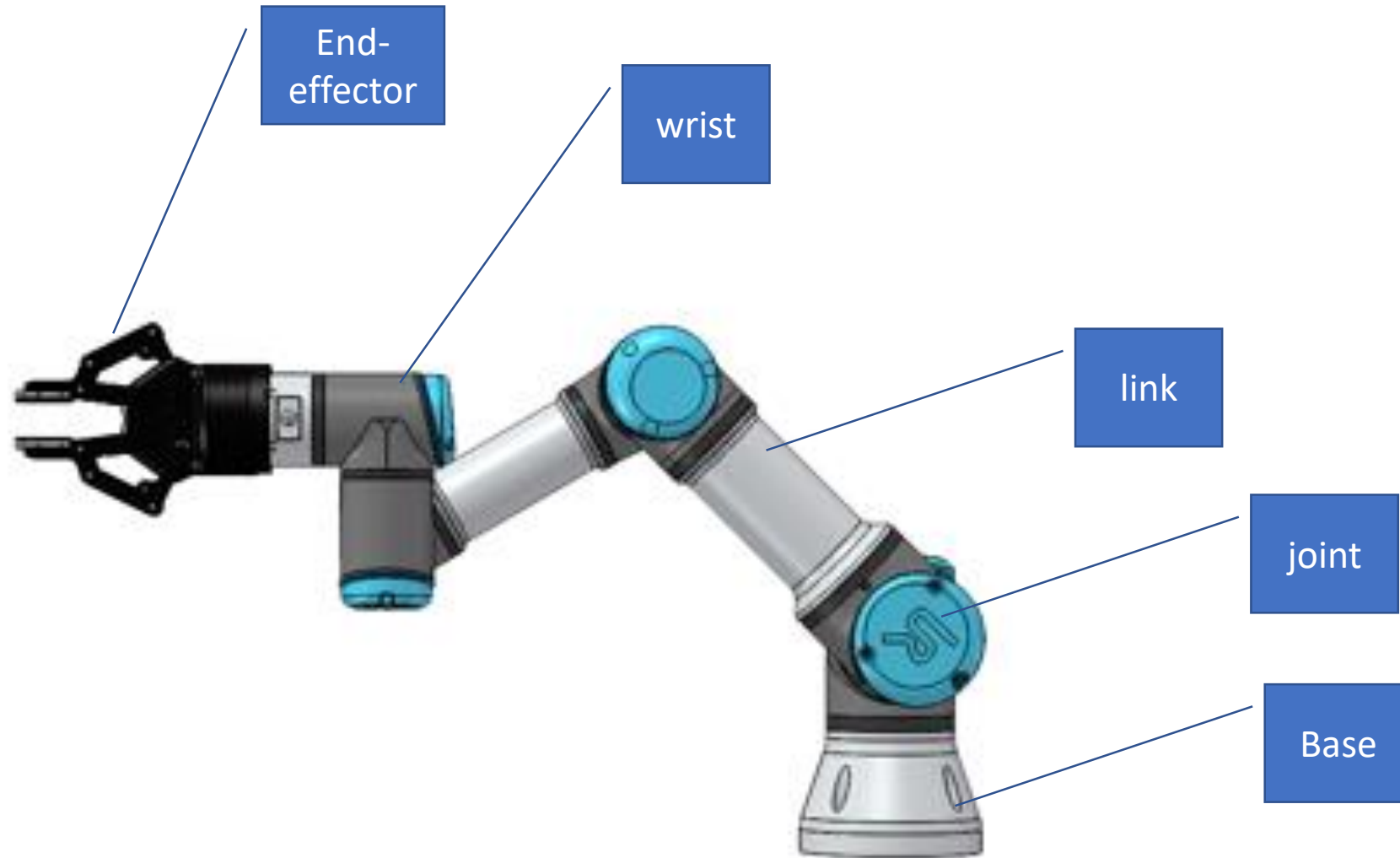
MOBILE ROBOTS: robots with a mobile base



Robotic Manipulators

- A robotic manipulator consists of
 - *An arm ensuring mobility*
 - A wrist located at the end of the arm enabling dextrous operation
 - An end-effector executing the robot's tasks
- From the geometric point of view, the manipulator is a sequence of rigid bodies (links), connected by mechanical articulations (joints).

Manipulator example

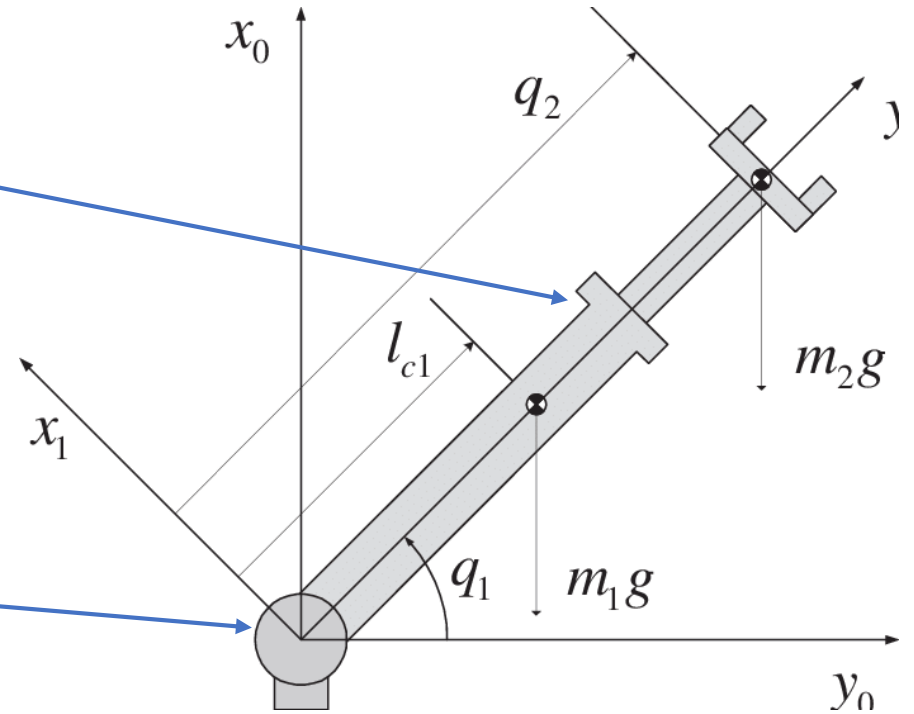


Joints

- Joints provide the structure with its necessary mobility
- Joints can be of two types

Prismatic joints: enable a relative translational motion between the links

Revolute joints: enable a relative rotational motion between the links

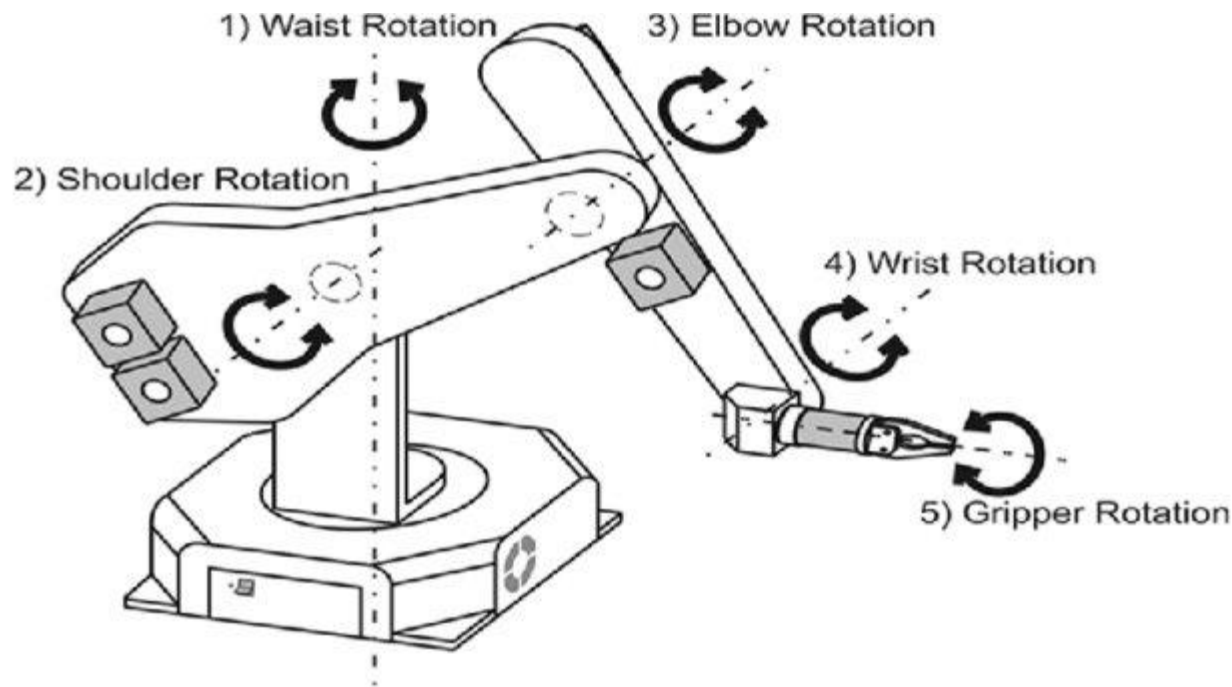


A few definitions

- **Dexterity** can be defined as a robot's ability to cope with a variety of objects and actions.
 - how robots can interact and handle objects and take necessary actions on the objects.
- **Stiffness:** ability of a body to resist deformation
 - In formal terms we can see the stiffness as the amount of force required to induce a motion along a DoF

Degrees of freedom

- In a mechanical structure a degree of freedom (DoF) defines a specific mode in which the robot can move
- Typically each joint is endowed with an actuator (e.g., a brushless, or a linear motor) and provides the structure with one degree of freedom
- The more the degrees of freedom the more flexible is the machine



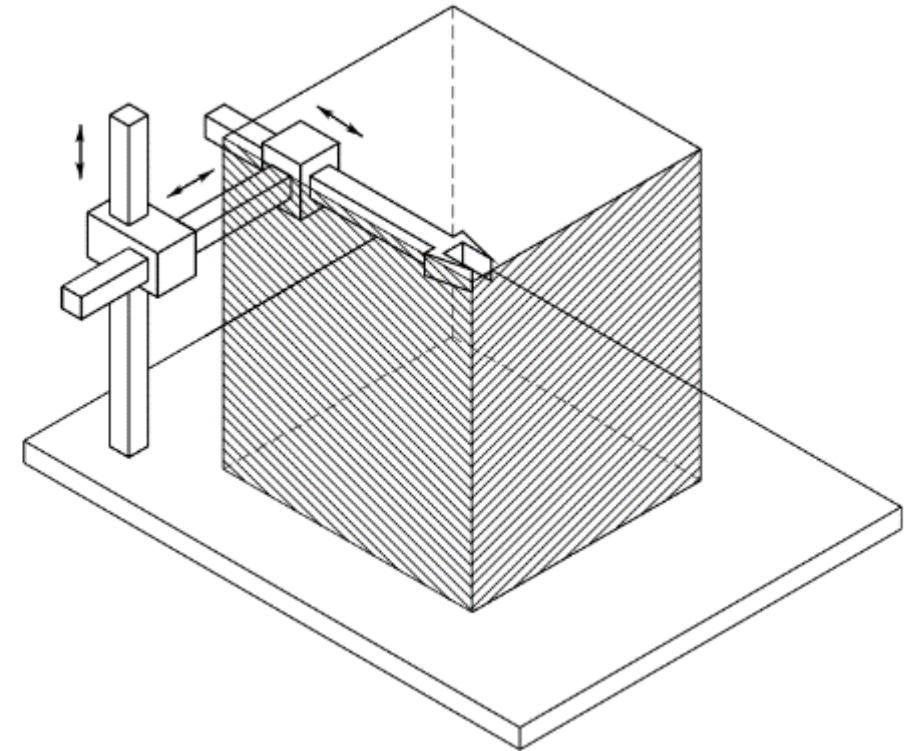
- The workspace is the area that the end effector can reach
- Its structure very much depends on the number and on the type of the joints

Different type of manipulators

- According to the structure of the different DoF we can distinguish between different types of manipulators
 - Cartesian
 - Cylindrical
 - Spherical
 - SCARA
 - Antropomorphic

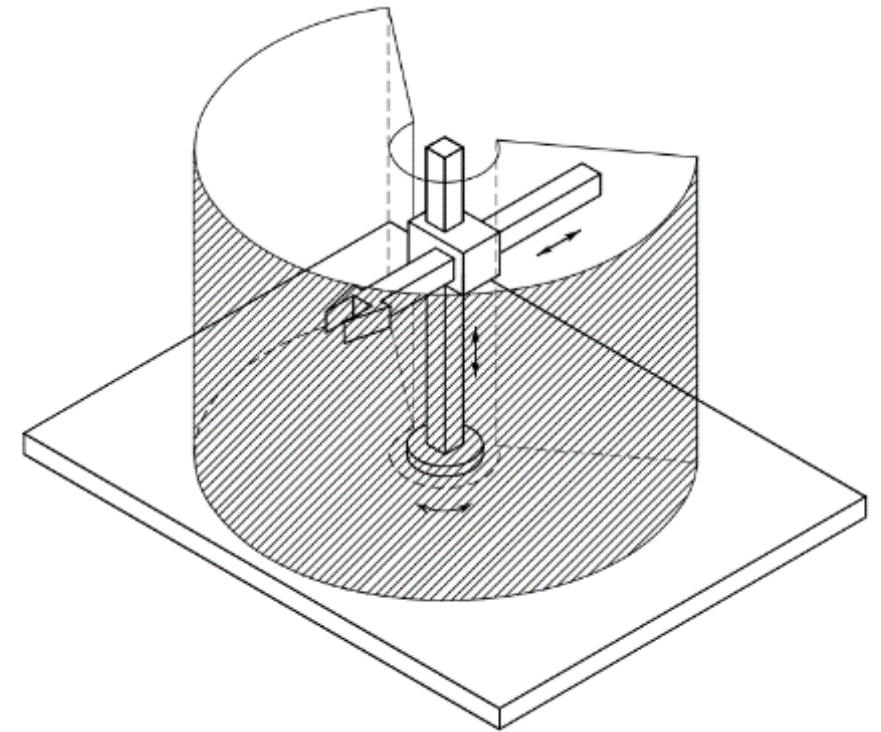
Cartesian

- Cartesian robots are characterised by three prismatic joints with three mutually orthogonal axis
- The workspace is parallelepiped
- IT operates with good mechanical stiffness and accuracy everywhere in its workspace
- The exclusive presence of prismatic joints reduces the structure's dexterity



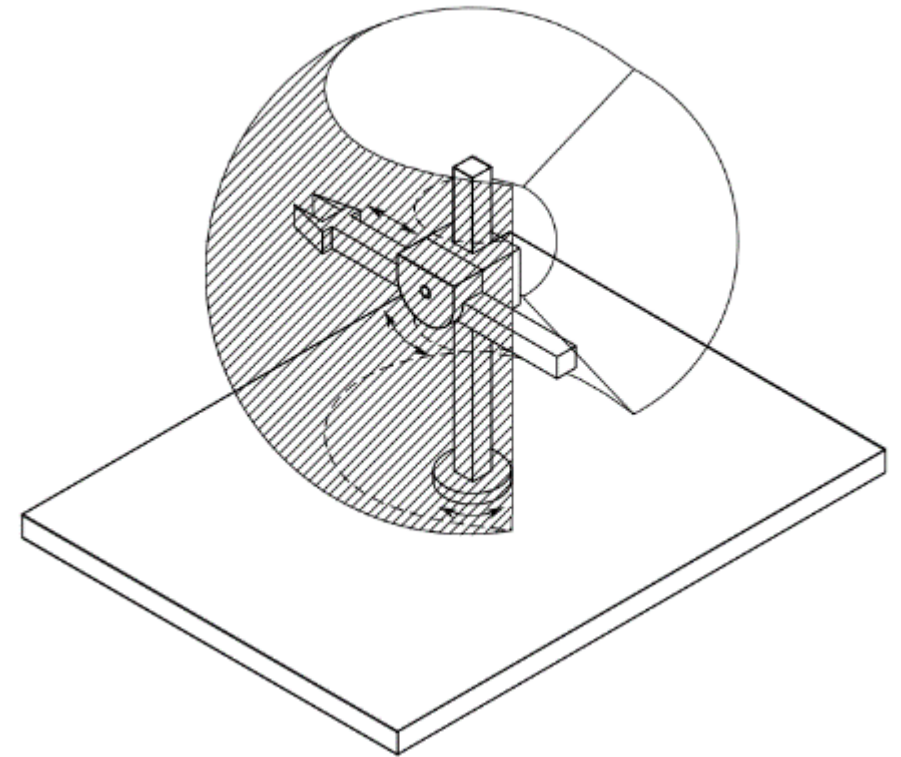
Cylindrical Manipulators

- In a cylindrical manipulators one of the joints is replaced by a revolute joints.
- Wrist accuracy decreases with horizontal stroke
- Good stiffness
- The workspace is a portion of cylinder
- This type of machine is primarily used to move heavy loads within cylindric cavities



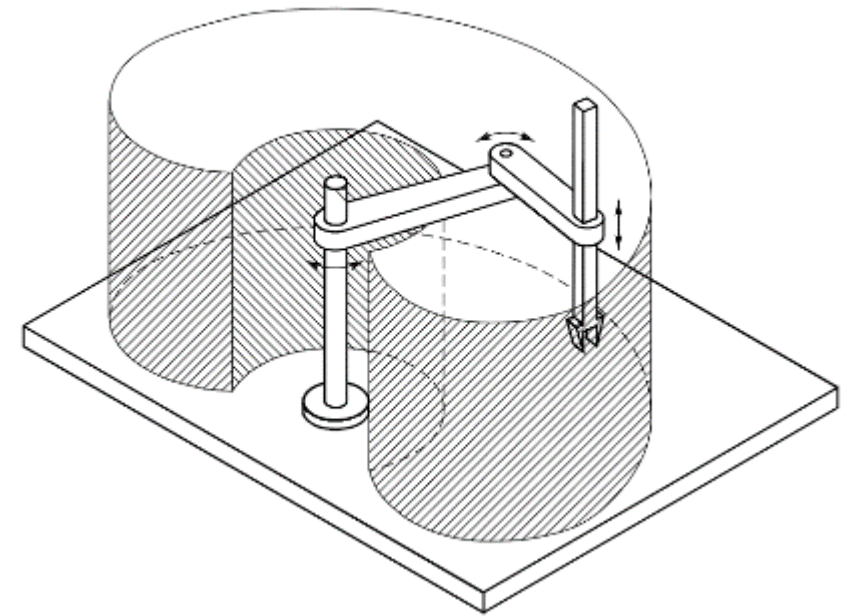
Spherical Manipulators

- For a spherical manipulator, we replace two prismatic joints with revolute joints
- The Workspace is a portion of a hollow sphere
- The manipulator has a reduced accuracy when the radial stroke increases
- Primarily used for machining

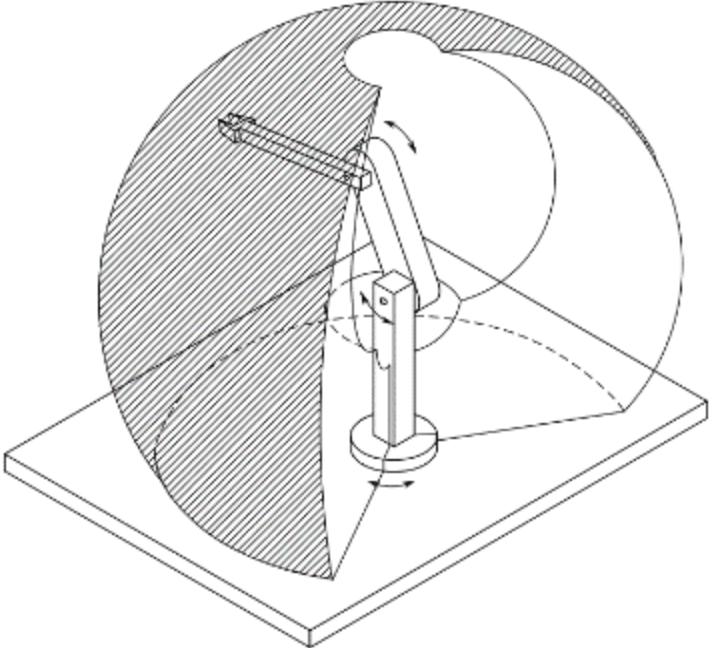


SCARA manipulators

- For a SCARA (**S**elective **C**ompliance **A**ssembly **R**obot **A**rm), we have two revolute joints and one prismatic joint
- The axis of motion of the SCARA are parallel
- High stiffness to vertical loads, compliance to horizontal loads
- Well suited to vertical assembly tasks and for the manipulation of small objects
- Positioning accuracy decreases with the distance from the first axis

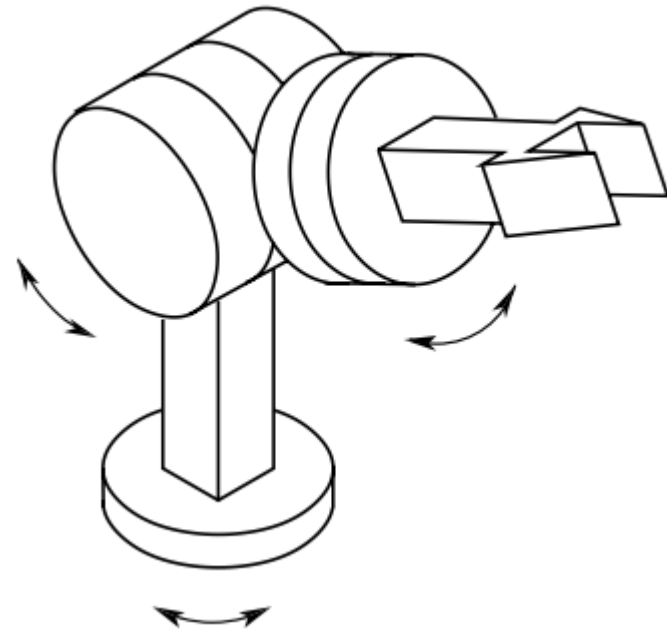


Anthropomorphic manipulators

- In the anthropomorphic manipulator we have three revolute joints.
 - The axis of the second and of the third joint are orthogonal to the axis of the first
 - The similarity with the human arm is evident
 - The second joint is said shoulder and the third is said elbow
- 
- The diagram illustrates an anthropomorphic manipulator with three revolute joints. It consists of a base plate, a vertical column, and a horizontal arm. The first joint is at the base, allowing rotation around a vertical axis. The second joint is at the top of the column, allowing rotation around a horizontal axis. The third joint is at the end of the arm, allowing rotation around a vertical axis. The workspace is shown as a shaded volume. Arrows indicate the direction of rotation for each joint.
- The three revolute joints maximise dexterity
 - The wrist position accuracy is different in the workspace
 - This robot is by far the most widespread in industry (59% of the installation)

Wrist

- All different manipulators have an end-effector attached to a wrist
- The end-effector is usually a gripper but can be a different thing according to the task
 - E.g., a welding gun, a spray gun, a screwdriver...
- The configuration of the wrist that maximises dexterity is spherical (three revolute joints)
- Each joint of a wrist provide an additional DoF

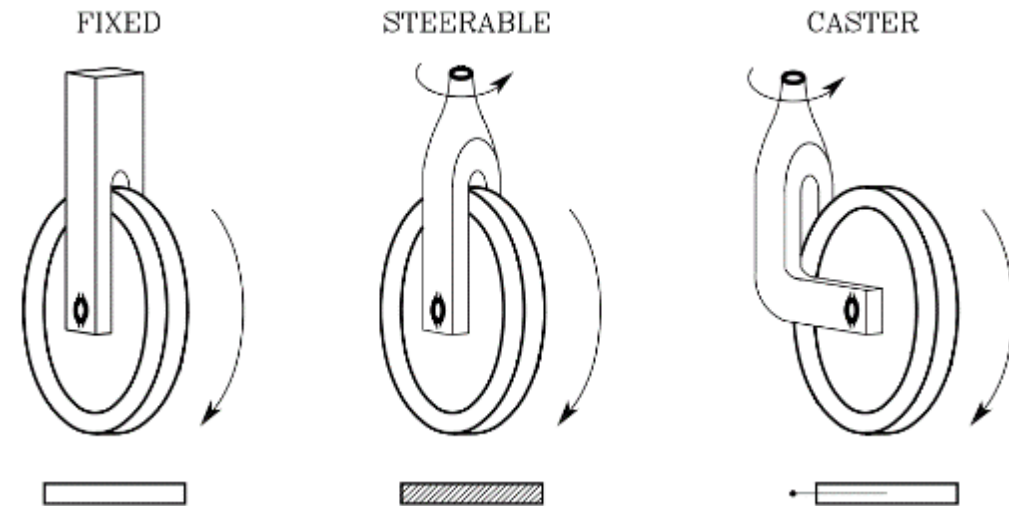


Mobil robot

- A mobile robot is characterised by a mobile base
- We can have two types of mobile robots
 - **Wheeled Robots:** A rigid body (chassis) and a system of wheels that provide motion with respect to the ground
 - Can be connected to a system of trailers connected by revolute joints
 - **Legged Robots:** Multiple rigid bodies connected through revolute (and sometimes) prismatic joints
 - Some of the links form lower limbs and feet that stay in contact with the ground and provide locomotion.
 - We will not consider legged robots in this course

Wheel types

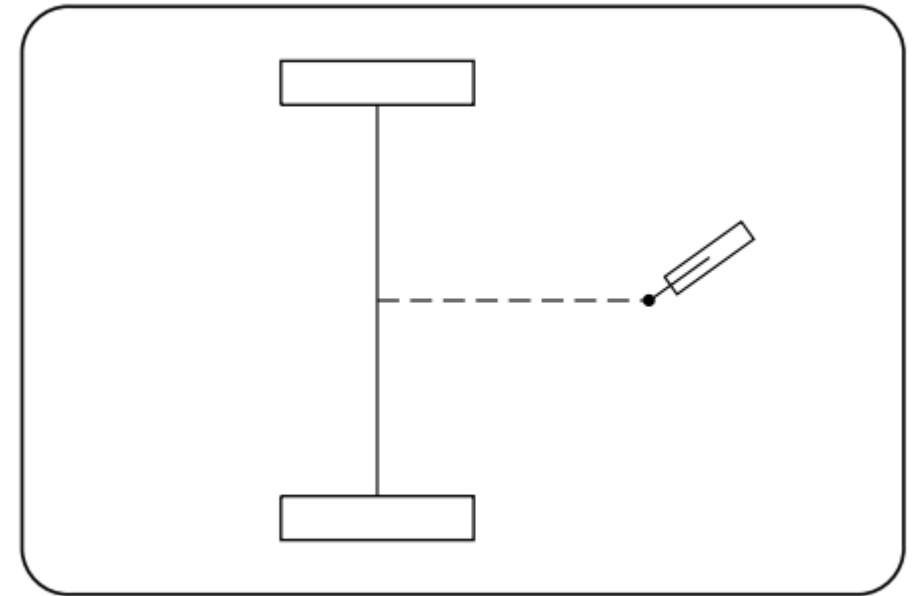
- Wheeled robots use three types of wheels
 - **Fixed:** can only rotate about an axis passing through the centre and orthogonal to the wheel plane
 - **Steerable:** has two axis of rotation: the standard axis typical of all wheels, a vertical axis passing through the centre
 - **Caster wheel:** a caster wheel is similar to a steerable wheel, but its vertical axis has an offset from the centre of the wheel.
 - The wheel swivels around when bending and automatically aligns itself with the direction of motion of the chassis



By combining different types of wheels we can obtain different types of kinematic structures

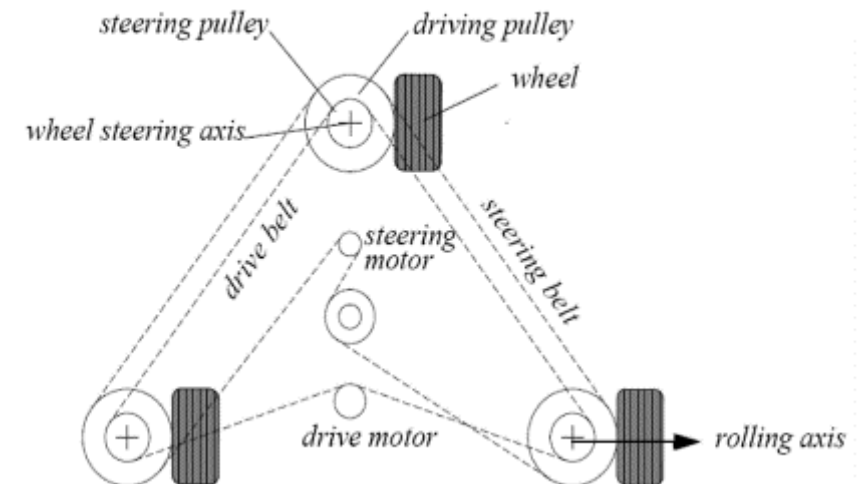
Differential drive vehicle

- In a differential drive robot there are
 - Two actuated fixed wheels with the same axis
 - A caster wheel to keep the robot statically balanced
 - The robot can rotate by applying a different velocity to the wheels, or move straight applying the same angular velocity
 - It can rotate on the spot by setting the two speeds to opposite values



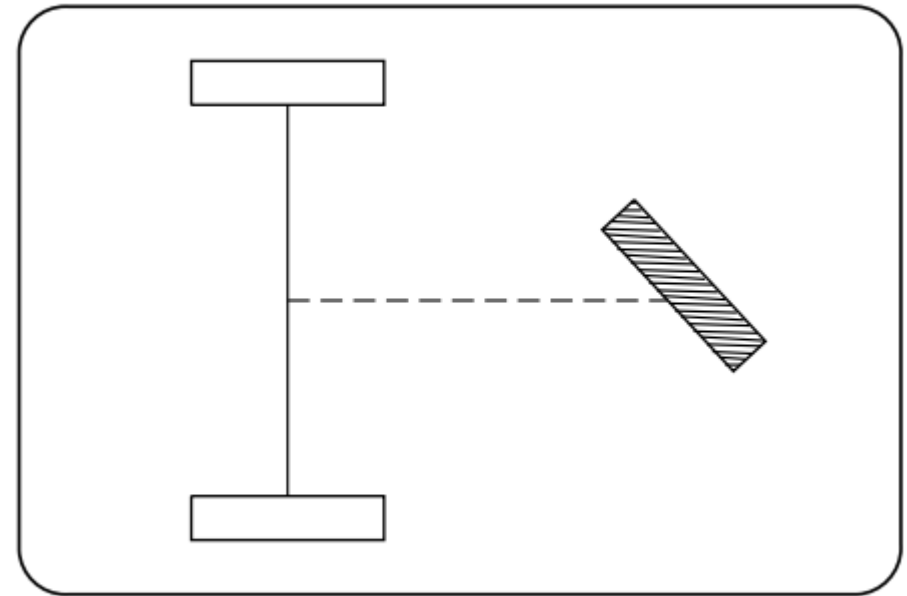
Synchro Drive robots

- In the synchro drive configuration there are three steerable wheels connected by a chain
- There are two motors: one rotates the three wheels “synchronously” and the other transmits the motion (synchronously) to all of them
- It can do the same kinematics of the differential drive but it needs only one motor to move straight



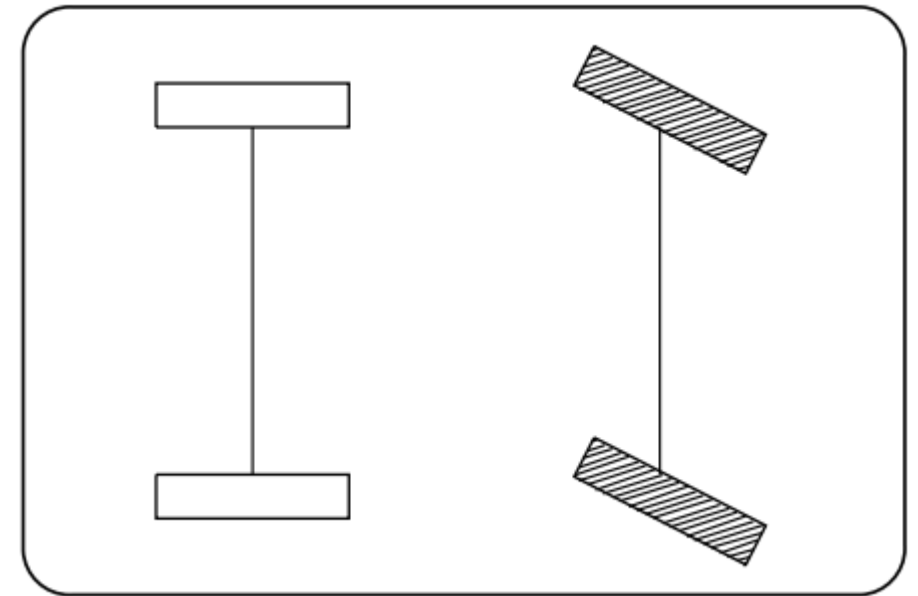
Tricycle robots

- In a tricycle we have two wheels that are fixed and actuated by a motor
- The third wheel is steerable and its rotation around the vertical axis is governed by another motor
- It is also possible that the two motors operate both on the steering wheel to turn it and to secure locomotion



Carlike

- The carlike robot is very similar to a tricycle except that it has two turning wheels
- Again the traction can be on the fixed wheels (rear-wheel traction) or on the front wheel (front-wheel traction)
- Contrary to a differential drive robot, a car-like (and a tricycle) cannot turn on the spot and have a limited curvature radius
- Besides, the carlike has an interesting problem

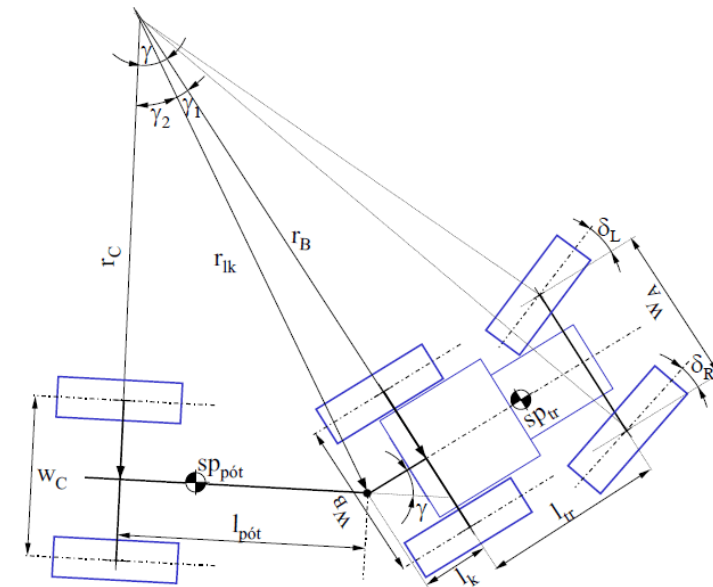
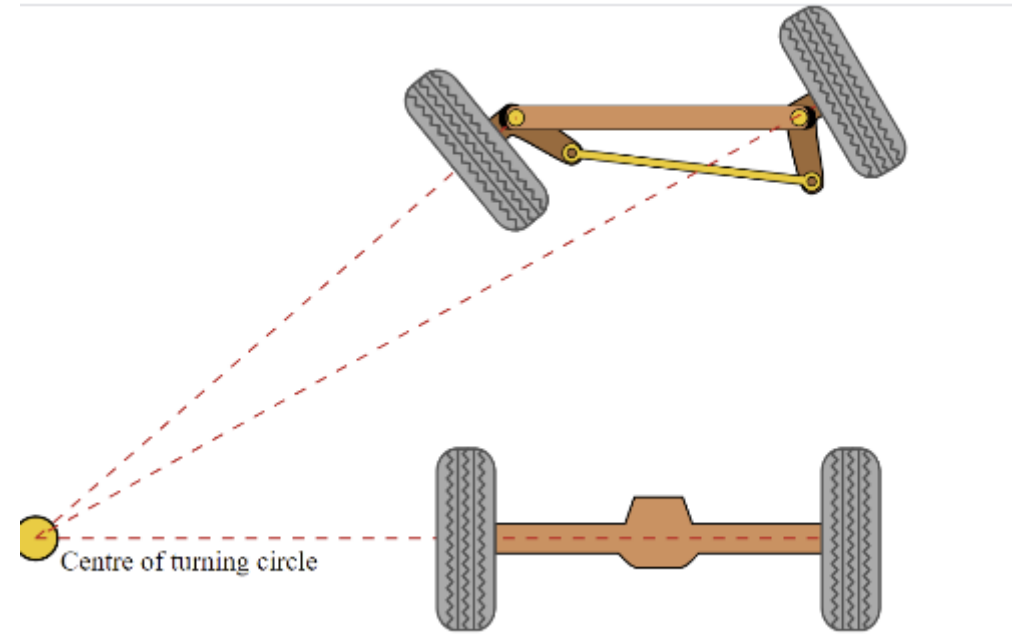


Ackerman steering

- Back in 1758 Erasmus Darwin was driving his horse-drawn carriage.
- The two front wheels turned of the same angle
- One of the wheels (the external one) slipped sideways and it caused the carriage to tip over his driver
- The solution was devised by Lankensperger in Munich and patented by his English agent (Ackerman) in London
- The idea is very simple: the two wheels have to follow the same circular trajectory

Ackerman Steering

- In order to follow the same circle the internal wheel has to turn more than the external one
- This is solved, in modern cars, by the geometry of the axle
- The Ackerman steering can be generalised to the case of trailers.

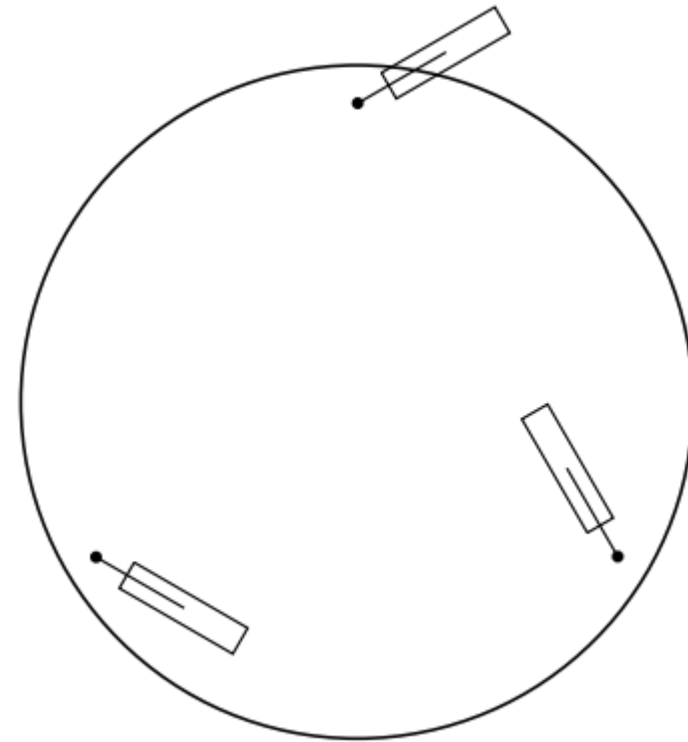


Constraints

- Contrary to manipulators mobile robots do not usually have a limited workspace
 - They can reach any location in the Euclidean space
- However they do have motion constraints
 - A Differential Drive robot is quite flexible, but it cannot move sideways along the axis connecting the actuated wheels
 - This is called a nonholonomic constraint
 - In addition, a carlike has a limited curvature radius

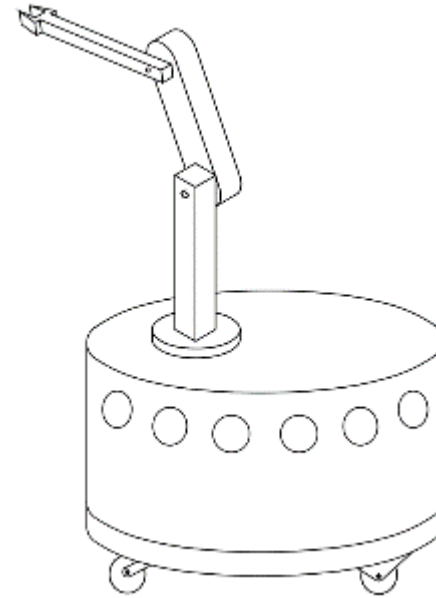
An exception

- An exception is an omnibot
- An omnibot is characterised by three autonomously actuated caster-wheels (usually in symmetric positions)
- It can move in *any direction*



Additional types of robots

- Clearly, we can mount an anthropomorphic manipulator and a differential drive mobile base in order to improve its working abilities
- We can improve the dexterity by additional degrees of freedom (7 for the Kuka LWR)



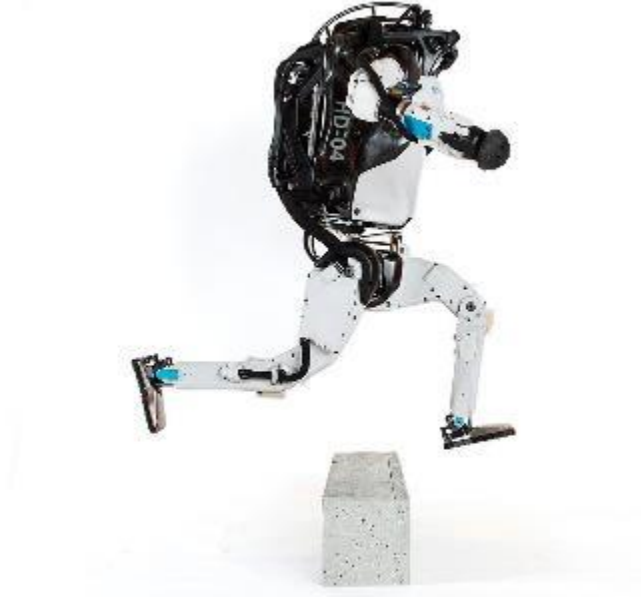
More sophisticated end-effectors

- A very active research area has been on the development of sophisticated anthropomorphic hands
- Such hands can be used for
 - rehabilitation and assistive purposes (e.g., as a replacement for a severed human hands)
 - Creation of sophisticated anthropomorphic robots



A new frontier: Humanoid Robots

- Putting together several types of robots we can create humanoid robots
-but this out of the scope of this course



Modelling: kinematics of manipulators

- **Kinematics** is about the motion of a robotic structure with respect to a generic frame
 - For a manipulator, given the joint position, find position and orientation of the end effector
- **Differential kinematics:**
 - For a manipulator: given the joint positions and motion (and their velocity) find the velocity of the end effector
- **Forward Kinematics:**
 - Find the end effector motion as a function of the joint motion
- **Inverse Kinematics**
 - Find the joint motion as a function of the “desired” end-effector motion and configuration

Modelling: kinematics of mobile robots

- Mobile robots:
 - The robot's kinematics requires a correct understanding of the motion constraints
 - The kinematic model translates the instantaneous motion of the motors into an instantaneous motion of the robot as a whole, which accounts for the robot's current configuration and for its motion constraints.